

Qualification Pack



Automotive Unified Diagnostics Engineer

QP Code: ASC/Q1437

Version: 1.0

NSQF Level: 6

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ASC/Q1437: Automotive Unified Diagnostics Engineer

Brief Job Description

The individual in this job roles performs diagnostic reverse engineering for all diagnostics protocol, diagnostic content coverage test and generate reports, vehicle tracing analysis and simulation, vehicle test for diagnostic content development and prepare diagnostic specification, design diagnostics workflow and testing report. The individual also buildup, Commission and Operation of the End2End Test benches, develop Test cases/execute for ECU/HIL system, Mobile app, API, Web services and handle all technical issues and ensure on time delivery of Test requests. He/she also performs feasibility check for Test Automation and Service Operation, Flash and protocol conformance validation of control units, test bench setup for ECUs Diagnostics testing and develop Test Strategy, Test Plan and test execution, test defects analysis and report, Automation tool scripting, verification, and validation.

Personal Attributes

The person should be result oriented with good technical and analytical skills, should have Excellent Interpersonal Skills, communication and presentation skills and a good team player. They should have ability to manage projects, prioritizing of work and mentoring the budding engineers

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ASC/N9810: Manage work and resources \(Manufacturing\)](#)
2. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)
3. [ASC/N1468: Carry out diagnosis of vehicle mechanical system](#)
4. [ASC/N1469: Carry out diagnosis of vehicle OBD system](#)
5. [ASC/N1470: Carry out diagnosis of vehicle electrical and electronic system](#)

Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
Country	India

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NSQF Level	6
Credits	21
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7213.0201
Minimum Educational Qualification & Experience	Completed 3 year diploma after 10th (Mechanical/Automobile/ Electrical / Electronics) with 5 Years of experience In relevant field OR B.E./B.Tech with 1 Year of experience In relevant field OR Certificate-NSQF (Four-wheeler Service Lead Technician Level 5) with 3 Years of experience in relevant field OR 10th Class + I.T.I (in engineering sector trades) with 5 Years of experience
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	Automotive Technician Level 5 Driving License
Minimum Job Entry Age	22 Years
Last Reviewed On	NA
Next Review Date	17/11/2025
NSQC Approval Date	17/11/2022
Version	1.0
Reference code on NQR	2022/AUT/ASDC/06448
NQR Version	1.1

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ASC/N9810: Manage work and resources (Manufacturing)

Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising the use of resources.

Scope

The scope covers the following :

- Maintain safe and secure working environment
- Maintain Health and Hygiene
- Effective waste management practices
- Material/energy conservation practices

Elements and Performance Criteria

Maintain safe and secure working environment

To be competent, the user/individual on the job must be able to:

- PC1.** identify hazardous activities and the possible causes of risks or accidents in the workplace
- PC2.** implement safe working practices for dealing with hazards to ensure safety of self and others
- PC3.** conduct regular checks of the machines with support of the maintenance team to identify potential hazards
- PC4.** ensure that all the tools/equipment/fasteners/spare parts are arranged as per specifications/utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/work instructions
- PC5.** organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices
- PC6.** fill daily check sheet to report improvements done and risks identified
- PC7.** ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others
- PC8.** report any identified breaches in health, safety and security policies and procedures to the designated person

Maintain Health and Hygiene

To be competent, the user/individual on the job must be able to:

- PC9.** ensure workplace, equipment, restrooms etc. are sanitized regularly
- PC10.** ensure team is aware about hygiene and sanitation regulations and following them on the shop floor
- PC11.** ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace
- PC12.** report advanced hygiene and sanitation issues to appropriate authority
- PC13.** follow stress and anxiety management techniques and support employees to cope with stress, anxiety etc
- PC14.** wear and dispose PPEs regularly and appropriately

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Effective waste management practices

To be competent, the user/individual on the job must be able to:

PC15. ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP

PC16. ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste

Material/energy conservation practices

To be competent, the user/individual on the job must be able to:

PC17. ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively

PC18. prepare and analyze material and energy audit reports to decipher excessive consumption of material and water

PC19. identify possibilities of using renewable energy and environment friendly fuels

PC20. identify processes where material and energy/electricity utilization can be optimized

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. organisation procedures for health, safety and security, individual role and responsibilities in this context

KU2. the organisation's emergency procedures for different emergency situations and the importance of following the same

KU3. evacuation procedures for workers and visitors

KU4. how and when to report hazards as well as the limits of responsibility for dealing with hazards

KU5. potential hazards, risks and threats based on the nature of work

KU6. various types of fire extinguisher

KU7. various types of safety signs and their meaning

KU8. appropriate first aid treatment relevant to different condition e.g. bleeding, minor burns, eye injuries etc.

KU9. relevant standards, procedures and policies related to 5S followed in the company

KU10. the various materials used and their storage norms

KU11. importance of efficient utilisation of material and water

KU12. basics of electricity and prevalent energy efficient devices

KU13. common practices of conserving electricity

KU14. common sources and ways to minimize pollution

KU15. categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics

KU16. waste management techniques

KU17. significance of greening

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** work with supervisors/team members to carry out work related tasks
- GS4.** complete tasks efficiently and accurately within stipulated time
- GS5.** inform/report to concerned person in case of any problem
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** write reports such as accident report, in at least English/regional language

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain safe and secure working environment</i>	20	13	-	8
PC1. identify hazardous activities and the possible causes of risks or accidents in the workplace	4	2	-	2
PC2. implement safe working practices for dealing with hazards to ensure safety of self and others	3	1	-	2
PC3. conduct regular checks of the machines with support of the maintenance team to identify potential hazards	2	2	-	1
PC4. ensure that all the tools/equipment/fasteners/spare parts are arranged as per specifications/utility into proper trays, cabinets, lockers as mentioned in the 5S guidelines/work instructions	3	2	-	1
PC5. organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices	2	-	-	-
PC6. fill daily check sheet to report improvements done and risks identified	2	2	-	-
PC7. ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others	2	2	-	1
PC8. report any identified breaches in health, safety and security policies and procedures to the designated person	2	2	-	1
<i>Maintain Health and Hygiene</i>	13	7	-	5
PC9. ensure workplace, equipment, restrooms etc. are sanitized regularly	3	2	-	1
PC10. ensure team is aware about hygiene and sanitation regulations and following them on the shop floor	2	1	-	-
PC11. ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace	2	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. report advanced hygiene and sanitation issues to appropriate authority	1	1	-	1
PC13. follow stress and anxiety management techniques and support employees to cope with stress, anxiety etc	2	1	-	1
PC14. wear and dispose PPEs regularly and appropriately	3	-	-	1
<i>Effective waste management practices</i>	6	4	-	1
PC15. ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP	3	2	-	-
PC16. ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste	3	2	-	1
<i>Material/energy conservation practices</i>	11	6	-	6
PC17. ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively	2	2	-	1
PC18. prepare and analyze material and energy audit reports to decipher excessive consumption of material and water	3	2	-	1
PC19. identify possibilities of using renewable energy and environment friendly fuels	3	1	-	2
PC20. identify processes where material and energy/electricity utilization can be optimized	3	1	-	2
NOS Total	50	30	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9810
NOS Name	Manage work and resources (Manufacturing)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	5
Credits	2
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values - Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment

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- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms

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- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services

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- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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ASC/N1468: Carry out diagnosis of vehicle mechanical system

Description

This NOS unit is about performing diagnosis of faults and failures in vehicle mechanical system/aggregates such as engine, clutch, gearbox, brakes, steering and suspension system etc. to identify their causes and preventive measures.

Scope

The scope covers the following :

- Prepare for diagnosis process
- Perform diagnosis of vehicle mechanical system/aggregates
- Perform post-diagnosis activities

Elements and Performance Criteria

Prepare for diagnosis process

To be competent, the user/individual on the job must be able to:

- PC1.** obtain the details of faults/failures in vehicle through field reports and DMS data received from dealership/OEM/Area office
- PC2.**
 - interpret the vehicle drawings, work instruction's, fault reports etc. to plan the diagnosis tasks need
 - to be conducted
- PC3.**
 - identify and select the diagnostic tools, testing tools, simulation tools, software testing tools, hand
 - tools, measuring tools, measuring instruments, gauges, etc. required during the diagnosis process
- PC4.**
 - ensure that tools, gauges and diagnosis apparatus are cleaned properly and free from any dust and
 - impurities
- PC5.** identify the testing parameters which need to be measured during the diagnosis procedure
- PC6.**
 - set the diagnostic apparatus and connect the various data capturing meters and gauges, data collection
 - systems and data loggers to capture the required failure data as per the diagnosis process and SOPs/WI
- PC7.**
 - dismantle and reassemble aggregates of vehicle for fault diagnosis as per the Drawing/Standard
 - Operating Procedure/Work Instruction

Perform diagnosis of vehicle mechanical system/aggregates

To be competent, the user/individual on the job must be able to:

- PC8.** perform root cause analysis to identify the cause of faults in the vehicle
- PC9.** carry out diagnosis of noise in engine
- PC10.** use compression tester and cylinder leakage tester to detect any leakage in cylinder

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- PC11.** use engine fault diagnosis table of symptom to identify possible causes and remedies for the identified faults/failures in engine
- PC12.** conduct cooling system leakage test, antifreeze content test and operating temperature test by using cooling fault diagnosis tables
- PC13.** conduct lubrication system diagnosis with oil pressure testing kit and by using lubrication fault diagnosis tables
- PC14.** conduct brake fluid diagnosis with brake fluid tester, brake roller test, disc runout test by using brake fluid diagnosis table
- PC15.** conduct steering and wheels diagnosis with tire pressure, tread depth, wheel balancer, four-wheel alignment machine by using steering fault diagnosis table and tire fault diagnostic table
- PC16.** conduct suspension diagnosis with damper operation test, trim height test by using suspension fault diagnosis table
- PC17.** conduct clutch diagnosis with clutch tests by using clutch fault diagnosis table
- PC18.** conduct automatic gearbox diagnosis with stall test and kickdown test using automatic gear box fault diagnosis table

Perform post-diagnosis activities

To be competent, the user/individual on the job must be able to:

- PC19.** record and collate all the results in the required formats
- PC20.** analyse and compare the results with the standard values, regulatory norms and benchmarked values to identify causes of faults and suggest remedies and preventing actions to repair or avoid them
- PC21.** submit the report to concerned department for rectification of faults in vehicle

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's diagnostic standards and processes
- KU2.** diagnostic terminology in automobile industry
- KU3.** how to write precise report
- KU4.** various repairing and diagnostic techniques for vehicle mechanical system
- KU5.** Root cause analysis and Fishbone diagram
- KU6.** different types of noises and possible source of them
- KU7.** difference between noise vibration and harshness
- KU8.** use of different mechanical gauges, tools, and instruments
- KU9.** working of different mechanical, hydraulic and pneumatic system
- KU10.** working of mechanicals in engine and its subsystems, clutch, transmission, and final drive, brakes, steering, and suspension system
- KU11.** different steering and suspension geometry angles
- KU12.** different tread wear pattern and causes of it
- KU13.** various national and international regulations, norms and standards on vehicles
- KU14.** how to read and interpret sketches and engineering drawings

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KU15. potential health and safety hazards and related safety precautions

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** work with supervisors/team members to carry out work related tasks
- GS4.** complete tasks efficiently and accurately within stipulated time
- GS5.** inform/report to concerned person in case of any problem
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** write reports such as accident report, in at least English/regional language

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for diagnosis process</i>	6	17	6	2
PC1. obtain the details of faults/failures in vehicle through field reports and DMS data received from dealership/OEM/Area office	1	3	1	-
PC2. - interpret the vehicle drawings, work instruction's, fault reports etc. to plan the diagnosis tasks need - to be conducted	1	3	-	1
PC3. identify and select the diagnostic tools, testing tools, simulation tools, software testing tools, hand tools, measuring tools, measuring instruments, gauges, etc. required during the diagnosis process	1	3	1	1
PC4. ensure that tools, gauges and diagnosis apparatus are cleaned properly and free from any dust and impurities	1	2	1	-
PC5. identify the testing parameters which need to be measured during the diagnosis procedure	1	2	1	-
PC6. - set the diagnostic apparatus and connect the various data capturing meters and gauges, data collection - systems and data loggers to capture the required failure data as per the diagnosis process and SOPs/WI	-	2	1	-
PC7. - dismantle and reassemble aggregates of vehicle for fault diagnosis as per the Drawing/Standard - Operating Procedure/Work Instruction	1	2	1	-
<i>Perform diagnosis of vehicle mechanical system/aggregates</i>	11	26	11	5
PC8. perform root cause analysis to identify the cause of faults in the vehicle	1	2	1	1
PC9. carry out diagnosis of noise in engine	1	2	1	1
PC10. use compression tester and cylinder leakage tester to detect any leakage in cylinder	1	3	1	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. use engine fault diagnosis table of symptom to identify possible causes and remedies for the identified faults/failures in engine	1	2	1	1
PC12. conduct cooling system leakage test, antifreeze content test and operating temperature test by using cooling fault diagnosis tables	1	2	1	1
PC13. conduct lubrication system diagnosis with oil pressure testing kit and by using lubrication fault diagnosis tables	1	2	1	-
PC14. conduct brake fluid diagnosis with brake fluid tester, brake roller test, disc runout test by using brake fluid diagnosis table	1	3	2	1
PC15. conduct steering and wheels diagnosis with tire pressure, tread depth, wheel balancer, four-wheel alignment machine by using steering fault diagnosis table and tire fault diagnostic table	1	3	-	-
PC16. conduct suspension diagnosis with damper operation test, trim height test by using suspension fault diagnosis table	1	2	1	-
PC17. conduct clutch diagnosis with clutch tests by using clutch fault diagnosis table	1	2	1	-
PC18. conduct automatic gearbox diagnosis with stall test and kickdown test using automatic gear box fault diagnosis table	1	3	1	-
<i>Perform post-diagnosis activities</i>	3	7	3	3
PC19. record and collate all the results in the required formats	1	2	1	1
PC20. analyse and compare the results with the standard values, regulatory norms and benchmarked values to identify causes of faults and suggest remedies and preventing actions to repair or avoid them	1	2	1	1
PC21. submit the report to concerned department for rectification of faults in vehicle	1	3	1	1
NOS Total	20	50	20	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1468
NOS Name	Carry out diagnosis of vehicle mechanical system
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	6
Credits	6
Version	1.0
Last Reviewed Date	17/11/2022
Next Review Date	17/11/2025
NSQC Clearance Date	17/11/2022

Qualification Pack

ASC/N1469: Carry out diagnosis of vehicle OBD system

Description

This NOS unit is about diagnosis of vehicle On Board Diagnostics (OBD) system used in modern vehicles for diagnosis of faults in it.

Scope

The scope covers the following :

- Prepare for diagnosis process
- Perform diagnosis of vehicle OBD system
- Perform post-diagnosis activities

Elements and Performance Criteria

Prepare for diagnosis process

To be competent, the user/individual on the job must be able to:

- PC1.** obtain the details of faults/failures in vehicle OBD system through field reports and DMS data received from dealership/OEM/Area office
- PC2.** interpret the vehicle drawings, work instruction's, fault reports etc. to plan the diagnosis tasks need to be conducted
- PC3.** identify the testing parameters which need to be measured during the diagnosis procedure
- PC4.** set the diagnostic apparatus and connect the various data capturing meters and gauges, data collection systems and data loggers to capture the required failure data as per the diagnosis process and SOPs/WI

Perform diagnosis of vehicle OBD system

To be competent, the user/individual on the job must be able to:

- PC5.** conduct diagnosis process on various sensors i.e., crank and cam sensor, wheel speed sensor, Throttle position sensor, Accelerator pedal position sensor, hot film air mass sensor, temperature sensor, knock sensor, oxygen/ lambda sensor, strain gauge, common rail fuel pressure sensor, MAP sensor, rain sensor, yaw rate and acceleration sensor etc. available in the vehicle
- PC6.** analyse and compare the signal of faulty sensor with the recommended input signal of correct sensor to identify the deviations in signal, cause and preventive measures of the fault in the sensor
- PC7.** conduct diagnosis process on various actuators i.e. rotary idle control valve, stepper motor, single point injector, multipoint injector, common rail diesel injector, carbon canister, glow plug etc. available in the vehicle
- PC8.** analyse and compare the signal of faulty actuator with the recommended signal of correct actuator to identify the deviations in signal, cause and preventive measures of the fault in the actuator
- PC9.** conduct diagnosis process on communication network systems i.e. CAN, LIN and FLEXRAY available in the vehicle

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- PC10.** analyse and compare the signal of faulty network system with the recommended signal of correct network signal to identify the deviations in signal, cause and preventive measures of the fault in the network system
- PC11.** record and collate all the results in the required formats
- PC12.** analyse and compare the results with the standard values, regulatory norms and benchmarked values to identify causes of faults and suggest remedies and preventing actions to repair or avoid them
- PC13.** submit the report to concerned department for rectification of faults in vehicle

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** system in system representation
- KU2.** open loop and closed loop system
- KU3.** different types of data sources used by OEMs
- KU4.** categories of different sensors used in modern automobile according to working principle
- KU5.** different types of generic Oscilloscopes used in Automotive market
- KU6.** different Oscilloscope graphical user interface
- KU7.** how to find correct result of sensors and actuator testing
- KU8.** how to find correct waveform of sensors and actuator testing
- KU9.** source code component implementation
- KU10.** vehicle emission control strategies
- KU11.** classification of Diagnostic Trouble Codes (DTC)
- KU12.** DTC code composition
- KU13.** features and technology of current systems
- KU14.** European Driving cycles, USA Driving cycles and Indian Driving Cycle
- KU15.** EURO Pollution norms and Indian Pollution norms
- KU16.** CAFE Norms
- KU17.** functioning of OBD3 and EOBD
- KU18.** pinout diagram of OBD2 connector
- KU19.** process of component monitoring
- KU20.** process of rationality testing
- KU21.** use of catalyst monitor and evaporative system monitor

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** work with supervisors/team members to carry out work related tasks

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- GS4.** complete tasks efficiently and accurately within stipulated time
- GS5.** inform/report to concerned person in case of any problem
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** write reports such as accident report, in at least English/regional language

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for diagnosis process</i>	4	19	5	2
PC1. obtain the details of faults/failures in vehicle OBD system through field reports and DMS data received from dealership/OEM/Area office	1	5	1	-
PC2. interpret the vehicle drawings, work instruction's, fault reports etc. to plan the diagnosis tasks need to be conducted	1	5	2	1
PC3. identify the testing parameters which need to be measured during the diagnosis procedure	1	5	2	1
PC4. set the diagnostic apparatus and connect the various data capturing meters and gauges, data collection systems and data loggers to capture the required failure data as per the diagnosis process and SOPs/WI	1	4	-	-
<i>Perform diagnosis of vehicle OBD system</i>	16	31	15	8
PC5. conduct diagnosis process on various sensors i.e., crank and cam sensor, wheel speed sensor, Throttle position sensor, Accelerator pedal position sensor, hot film air mass sensor, temperature sensor, knock sensor, oxygen/ lambda sensor, strain gauge, common rail fuel pressure sensor, MAP sensor, rain sensor, yaw rate and acceleration sensor etc. available in the vehicle	2	3	2	-
PC6. analyse and compare the signal of faulty sensor with the recommended input signal of correct sensor to identify the deviations in signal, cause and preventive measures of the fault in the sensor	2	3	2	1
PC7. conduct diagnosis process on various actuators i.e. rotary idle control valve, stepper motor, single point injector, multipoint injector, common rail diesel injector, carbon canister, glow plug etc. available in the vehicle	2	4	2	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. analyse and compare the signal of faulty actuator with the recommended signal of correct actuator to identify the deviations in signal, cause and preventive measures of the fault in the actuator	1	3	2	1
PC9. conduct diagnosis process on communication network systems i.e. CAN, LIN and FLEXRAY available in the vehicle	2	3	-	1
PC10. analyse and compare the signal of faulty network system with the recommended signal of correct network signal to identify the deviations in signal, cause and preventive measures of the fault in the network system	2	4	2	1
PC11. record and collate all the results in the required formats	1	3	2	1
PC12. analyse and compare the results with the standard values, regulatory norms and benchmarked values to identify causes of faults and suggest remedies and preventing actions to repair or avoid them	2	4	2	1
PC13. submit the report to concerned department for rectification of faults in vehicle	2	4	1	1
NOS Total	20	50	20	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1469
NOS Name	Carry out diagnosis of vehicle OBD system
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	6
Credits	4
Version	1.0
Last Reviewed Date	17/11/2022
Next Review Date	17/11/2025
NSQC Clearance Date	17/11/2022

Qualification Pack

ASC/N1470: Carry out diagnosis of vehicle electrical and electronic system

Description

This NOS unit is about performing diagnosis of faults and failures in vehicle electrical and electronic system/aggregates such as lighting system, ABS, HVAC system etc. to identify their causes and preventive measures.

Scope

The scope covers the following :

- Prepare for diagnosis process
- Perform diagnosis of vehicle electrical and electronic system/aggregates
- Perform post-diagnosis activities

Elements and Performance Criteria

Prepare for diagnosis process

To be competent, the user/individual on the job must be able to:

- PC1.** obtain the details of faults/failures in vehicle OBD system through field reports and DMS data received from dealership/OEM/Area office
- PC2.** interpret the vehicle drawings, wiring diagrams, work instruction's, fault reports etc. to plan the diagnosis tasks need to be conducted
- PC3.** identify the testing parameters which need to be measured during the diagnosis procedure
- PC4.** identify color codes and terminal numbers of the wires through wiring diagram
- PC5.** set the diagnostic apparatus and connect the various data capturing meters and gauges, data collection systems and data loggers to capture the required failure data as per the diagnosis process and SOPs/WI

Perform diagnosis of vehicle electrical system/aggregates

To be competent, the user/individual on the job must be able to:

- PC6.** use analogue meters or multimeter to check voltage, resistance and current in the vehicle circuit and test lighting system of the vehicle
- PC7.** diagnose lighting system fault by using lighting system diagnosis chart
- PC8.** diagnose auxiliaries' system fault by using auxiliaries' system diagnosis chart
- PC9.** diagnose security and communication system fault with security and communication system diagnosis chart
- PC10.** diagnose body electrical system fault with body electrical system diagnosis chart
- PC11.** diagnose HVAC system fault with HVAC system diagnosis chart
- PC12.** diagnose SRS system fault with SRS system diagnosis chart
- PC13.** check flashing and calibration changes in ECU of the vehicle

Perform diagnosis of vehicle electronic system/aggregates

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC14.** use available scanner and exhaust gas analyzer to diagnose fuel system by using fuel fault
- PC15.** use oscilloscope to diagnose ignition system by using ignition fault diagnosis table
- PC16.** use available scanner, smoke meter and gas analyzer to diagnose fuel injection system by using fuel injection fault diagnosis table
- PC17.** construct component aspect that provides more detailed description of the timing behavior of atomic software component
- PC18.** check health of battery health by using multimeter, hydrometer, and intelligent battery testers
- PC19.** diagnose starting system fault with starting system diagnosis chart
- PC20.** diagnose charging system fault with charging system diagnosis chart
- PC21.** diagnose ESP system fault with ESP system diagnosis chart
- PC22.** use suitable scanner to diagnose Active suspension with diagnostic chart

Perform post-diagnosis activities

To be competent, the user/individual on the job must be able to:

- PC23.** record and collate all the results in the required formats
- PC24.** analyse and compare the results with the standard values, regulatory norms and benchmarked values to identify causes of faults and suggest remedies and preventing actions to repair or avoid them
- PC25.** submit the report to concerned department for rectification of faults in vehicle

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** process of voltage drop test and short circuit to earth test
- KU2.** how to measure voltage, current and resistance
- KU3.** components and functioning of various electrical and electronic systems of vehicle
- KU4.** different conditions of spark plug according to the symptoms
- KU5.** working of glove plug system
- KU6.** functioning of closed loop lambda system
- KU7.** air intake systems and exhaust system for Gasoline and Diesel
- KU8.** type of batteries and its specification
- KU9.** components and working of Starter and charging circuits
- KU10.** gasoline engine management fault finding chart
- KU11.** different battery charging methods
- KU12.** how to read wiring diagram
- KU13.** symbols used in wiring diagram
- KU14.** different bus systems such as drive train, multimedia, ADAS, Body bus etc

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** read instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** work with supervisors/team members to carry out work related tasks
- GS4.** complete tasks efficiently and accurately within stipulated time
- GS5.** inform/report to concerned person in case of any problem
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** write reports such as accident report, in at least English/regional language

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for diagnosis process</i>	5	15	5	2
PC1. obtain the details of faults/failures in vehicle OBD system through field reports and DMS data received from dealership/OEM/Area office	1	3	1	-
PC2. interpret the vehicle drawings, wiring diagrams, work instruction's, fault reports etc. to plan the diagnosis tasks need to be conducted	1	3	1	1
PC3. identify the testing parameters which need to be measured during the diagnosis procedure	1	3	1	1
PC4. identify color codes and terminal numbers of the wires through wiring diagram	1	3	1	-
PC5. set the diagnostic apparatus and connect the various data capturing meters and gauges, data collection systems and data loggers to capture the required failure data as per the diagnosis process and SOPs/WI	1	3	1	-
<i>Perform diagnosis of vehicle electrical system/aggregates</i>	8	18	9	4
PC6. use analogue meters or multimeter to check voltage, resistance and current in the vehicle circuit and test lighting system of the vehicle	1	2	2	-
PC7. diagnose lighting system fault by using lighting system diagnosis chart	1	2	1	-
PC8. diagnose auxiliaries' system fault by using auxiliaries' system diagnosis chart	1	2	1	1
PC9. diagnose security and communication system fault with security and communication system diagnosis chart	1	3	1	1
PC10. diagnose body electrical system fault with body electrical system diagnosis chart	1	2	1	-
PC11. diagnose HVAC system fault with HVAC system diagnosis chart	1	2	1	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. diagnose SRS system fault with SRS system diagnosis chart	1	2	1	1
PC13. check flashing and calibration changes in ECU of the vehicle	1	3	1	-
<i>Perform diagnosis of vehicle electronic system/aggregates</i>	6	14	6	3
PC14. use available scanner and exhaust gas analyzer to diagnose fuel system by using fuel fault	1	2	1	-
PC15. use oscilloscope to diagnose ignition system by using ignition fault diagnosis table	1	3	1	1
PC16. use available scanner, smoke meter and gas analyzer to diagnose fuel injection system by using fuel injection fault diagnosis table	1	2	1	-
PC17. construct component aspect that provides more detailed description of the timing behavior of atomic software component	1	2	1	-
PC18. check health of battery health by using multimeter, hydrometer, and intelligent battery testers	-	1	-	1
PC19. diagnose starting system fault with starting system diagnosis chart	1	1	-	-
PC20. diagnose charging system fault with charging system diagnosis chart	-	1	1	1
PC21. diagnose ESP system fault with ESP system diagnosis chart	1	1	1	-
PC22. use suitable scanner to diagnose Active suspension with diagnostic chart	-	1	-	-
<i>Perform post-diagnosis activities</i>	1	3	-	1
PC23. record and collate all the results in the required formats	-	1	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. analyse and compare the results with the standard values, regulatory norms and benchmarked values to identify causes of faults and suggest remedies and preventing actions to repair or avoid them	1	1	-	-
PC25. submit the report to concerned department for rectification of faults in vehicle	-	1	-	-
NOS Total	20	50	20	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1470
NOS Name	Carry out diagnosis of vehicle electrical and electronic system
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	6
Credits	6
Version	1.0
Last Reviewed Date	17/11/2022
Next Review Date	17/11/2025
NSQC Clearance Date	17/11/2022

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

Qualification Pack

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N9810.Manage work and resources (Manufacturing)	50	30	-	20	100	15
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	10
ASC/N1468.Carry out diagnosis of vehicle mechanical system	20	50	20	10	100	25
ASC/N1469.Carry out diagnosis of vehicle OBD system	20	50	20	10	100	25
ASC/N1470.Carry out diagnosis of vehicle electrical and electronic system	20	50	20	10	100	25
Total	130	210	60	50	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.